

Sr. No.		Description	UOM (Wherever Applicable)	Data (Common For All Models)
A		General Points		
	1	Cooling Capacity	ton _R	Refer KCPL Chiller Selection System Software
	2	Power Consumption	kW	Refer KCPL Chiller Selection System Software
	3	Specific Power Consumption	kW/ton _R	Refer KCPL Chiller Selection System Software
	4	Co-Efficient of Performance (COP)	kW/kW	Refer KCPL Chiller Selection System Software
	5	No. of Compressors	Nos.	Contact To Engineering Team
	6	No. of Individual Refrigerant Circuits	Nos.	Contact To Engineering Team
	7	Refrigerant		
	i	Name	-	R134a
	ii	Quantity	kg	Refer KCPL Chiller Selection System Software
	iii	Technical Specifications	-	Refer ESP-18-19-003
	8	Sound Pressure Level		
	i	Noise Level	dB	Consult with Engineering Department on Case to Case Basis
	ii	Measuring Standard	-	ANSI/AHRI Standard 575-2008
	9	Insulation Details		
	i	Material	-	Closed Cell Nitrile Foam
	ii	Insulation Thickness on Various Parts	-	For Standard Temperature Range (LWT upto 3 °C)
		Evaporator Shell	mm	32
		Evaporator Tubesheet	mm	19
		Evaporator Dished End	mm	19
		Evaporator M.W.Box (If Applicable)	mm	19
		Evaporator Support Plate	mm	19
		Compressor Motor Body	mm	19
		Suction Line Assembly	mm	19
		Liquid Line Assembly	mm	9
	iii	Density	kg/m ³	76.6
	iv	Thermal Conductivity	W/m.K	0.035 (at 0 °C Mean Temperature)
	v	Standard	-	IS 14164
	vi	Adhesive	-	Blend of Synthetic Polymers and Synthetic Resin
	vii	Insulation Specifications	-	Refer ESP-18-19-004
	10	Vibration		
	i	Vibration Level	mm/sec	Less than 1.5 mm/sec
	ii	Vibration control	-	Rubber Pads (Standard) / Spring Isolators (At an Additional Cost)
	iii	Standard	-	IS 12075
	11	Painting Specification		
	i	Paint Type	-	RAL 7035
	ii	Standard	-	Coating as per KCPL Standards
	12	Overall Dimensions		
	i	Approx. Length	mm	Refer KCPL Chiller Selection System Software
	ii	Approx. Width	mm	Refer KCPL Chiller Selection System Software
	iii	Approx. Height	mm	Refer KCPL Chiller Selection System Software
	13	Space Clearances Required		
	i	Plain End Side (For Tube Cleaning)	mm	Contact To Engineering Team
	ii	All Other Sides	mm	Contact To Engineering Team
	iii	Overhead	mm	Contact To Engineering Team
	14	Weight		
	i	Approx. Shipping Weight	kg	Refer KCPL Chiller Selection System Software
	ii	Approx. Operating Weight	kg	Refer KCPL Chiller Selection System Software
	15	Cable Sizes		
	i	Aluminum Cable	-	Refer ESP-14-15-01
	ii	Copper Cable	-	Refer ESP-14-15-01
B		Compressor Details		
	1	Make	-	Refer "Make List" Sheet
	2	Type / Description	-	Semi Hermetic, Magnetic Bearings Centrifugal Compressor
	3	Model	-	Refer KCPL Chiller Selection System Software
	4	Drive	-	Direct Driven by Rotor Shaft
	5	Compressor Speed (Impeller Tip Speed)	RPM	Consult with Engineering Department on Case to Case Basis
	6	Capacity Control Percentage	%	Refer KCPL Chiller Selection System Software
	7	Type of Capacity Control	-	Stepless
	8	Capacity Control Mechanism	-	IGV and Speed Variation
	9	Design and Test Parameters		
	i	Design Pressure	bar	Consult with Engineering Department on Case to Case Basis
	ii	Test Pressure	kgf/cm ²	Consult with Engineering Department on Case to Case Basis
	iii	Design Temperature	°C	Consult with Engineering Department on Case to Case Basis
	iv	Max. Allowable Discharge Temperature	°C	Consult with Engineering Department on Case to Case Basis

	10		Bearings		
		i	Types of Bearings	-	Magnetic Bearings
	11		Lubrication	-	Being Mag-Bearing Compressor, No Lubication is Required
	12		Compressor Components MOC		
		i	Impeller	-	Proprietary Data
		ii	Casing	-	Proprietary Data
		iii	Shaft	-	Proprietary Data
	13		Physical Data of Compressor		
		i	Impeller Diameter	mm	Consult with Engineering Department on Case to Case Basis
		ii	No. of Impeller Stages	Nos.	Consult with Engineering Department on Case to Case Basis
	14		Copressor Isolation Type		
		i	At Suction	-	Butterfly Valve
		ii	At Discharge	-	Check Valve (NRV)
C			Compressor Motor Details		
	1		Make	-	Refer "Make List" Sheet
	2		Type of Duty	-	Continuous
	3		Motor Rating	kW	Refer KCPL Chiller Selection System Software
	4		Motor Speed	RPM	Consult with Engineering Department on Case to Case Basis
	5		Performance Indicators		
		i	Motor Efficiency Class	-	NA
		ii	Motor Power	kW	Refer KCPL Chiller Selection System Software
	6		Motor Cooling		
		i	Motor Cooling Type	-	Refrigerant Cooled
		ii	Cooling Mechanism	-	Liquid Refrigerant
		iii	Temperature at full load	⁰ C	10 to 15 (At Normal Condtions)
	7		Current Details		
		ii	Rated Load Current	A	Refer KCPL Chiller Selection System Software
		iii	Full Load Current	A	Refer KCPL Chiller Selection System Software
		iv	Inrush/Starting Current	A	Refer KCPL Chiller Selection System Software
		v	Locked Rotor Current	A	Refer KCPL Chiller Selection System Software
	8		Control Settings		
		i	No. of Starts per Hour	Nos.	6
		ii	Time Between STOP to START	Sec	300
		iii	Time Between START to START	Sec	600
D			Power Supply (Standard-Chiller Icomer)		
	1		Supply Voltage	V	415
	2		Permissible Voltage Variation	%	±10%
	3		Frequency	Hz	50
	4		Permissible Frequency Variation	%	±3%
	5		Phase	-	3
	6		Control Voltage	V	230 (Standard)
					110 (Special-Optional)
	7		Supply Wire System	-	3 Phase - 4 Wire System (Standard)
					3 Phase - 3 Wire System (Special-Optional)
	8		Fault Level at Busbar	kA	As per KCPL Standard Practice
E			Evaporator Details		
	1		Model	-	Refer KCPL Chiller Selection System Software
	2		Design Code	-	As per KCPL Standards
	3		Type	-	Shell and Tube Flooded Design
	4		Tube Side (Fluid)	-	Chilled Water
	5		Shell Side (Fluid)	-	Refrigerant
	6		Design Parameters		
		i	Design Temperature (Refrigerant Side)	⁰ C	65
		ii	Max. Operating Pressure (Refrigerant Side)	bar	Refer ESP-07-08-107
		iii	Design Pressure (Refrigerant Side)	bar	Refer ESP-07-08-107
		iv	Test pressure (Refrigerant Side)	bar	Refer ESP-07-08-107
		v	Testing method (Refrigerant Side)	-	Refer ESP-07-08-107
		vi	No. of Passes (Refrigerant Side)	Nos.	Single Pass
		vii	Design Temperature (Water Side)	⁰ C	65
		viii	Max. Operating Pressure (Water Side)	bar	Refer ESP-07-08-107
		ix	Design Pressure (Water Side)	bar	Refer ESP-07-08-107
		x	Test pressure (Water Side)	bar	Refer ESP-07-08-107
		xi	Testing method (Water Side)	-	Refer ESP-07-08-107
		xii	No. of Passes (Water Side)	Nos.	Two Pass
		xiii	Water Velocity	m/s	Less than 3 m/s
		xiv	Inlet Pressure (Water Side)	bar	Depends on Site Piping Layout (Maximum Allowable - 9.4 bar)
		xv	Evaporating Temperature	⁰ C	Consult with Engineering Department on Case to Case Basis
	7		Physical Data of Evaporator		
		i	Overall Length of Evaporator	ft	Contact To Engineering Team
		ii	Shell Diameter	inch	Contact To Engineering Team

		iii	Shell Thickness	mm	Contact To Engineering Team
		iv	Approx. Shell Length	mm	Contact To Engineering Team
		v	Material of Construction of Shell	-	Mild Steel
		vi	Material Standard of Shell	-	Refer "MOC" Sheet
		vii	Tube Type/ Nature of Tube Surface	-	Integral Helical Fins on the Outside Surface and Integral Helical Ridges on the Inside Surface
		viii	Tube Length	mm	Refer "HX Details" Sheet
		ix	Tube Diameter	mm	Refer "HX Details" Sheet
		x	Tube Thickness	mm	Refer "HX Details" Sheet
		xi	Material of Construction of Tube	-	Cu
		xii	Material Standard of Tube	-	Refer "MOC" Sheet
		xiii	Water Volume in Evaporator	Liter	Refer KCPL Chiller Selection System Software
	8		Water Box Details		
		i	Type	-	Standard - Dish Ends (M.W.Box - Optional)
		ii	Material	-	Mild Steel
		iii	Material Standard	-	Refer "MOC" Sheet
		iv	Nozzle size	NB	Refer KCPL Chiller Selection System Software
		v	End connection	-	Standard - Victaulic Conn. (Flanged Conn. - Optional)
		vi	MOC of Water Side Gasket	-	NAM AF 120
		vii	MOC of Refrigerant Side Gasket	-	NAM AF 159
	9		Accessories Provided		
		i	Pressure Relief Valve	-	Spring Loaded (For Safety Valve Set Pressure Refer ESP)
		ii	Drain/Vent Valves	Inch	Plugged Connection Provided (3/8" NPT)
F			Condenser Details		
	1		Model	-	Refer KCPL Chiller Selection System Software
	2		Design Code	-	As per KCPL Standards
	3		Type	-	Shell and Tube Flooded Design
	4		Tube Side (Fluid)	-	Chilled Water
	5		Shell Side (Fluid)	-	Refrigerant
	6		Design Parameters		
		i	Design Temperature (Refrigerant Side)	⁰ C	100
		ii	Max. Operating Pressure (Refrigerant Side)	bar	Refer ESP-07-08-107
		iii	Design Pressure (Refrigerant Side)	bar	Refer ESP-07-08-107
		iv	Test pressure (Refrigerant Side)	bar	Refer ESP-07-08-107
		v	Testing method (Refrigerant Side)	-	Refer ESP-07-08-107
		vi	No. of Passes (Refrigerant Side)	Nos.	Single Pass
		vii	Design Temperature (Water Side)	⁰ C	100
		viii	Max. Operating Pressure (Water Side)	bar	Refer ESP-07-08-107
		ix	Design Pressure (Water Side)	bar	Refer ESP-07-08-107
		x	Test pressure (Water Side)	bar	Refer ESP-07-08-107
		xi	Testing method (Water Side)	-	Refer ESP-07-08-107
		xii	No. of Passes (Water Side)	Nos.	Two Pass
		xiii	Water Velocity	m/s	Less than 3 m/s
		xiv	Inlet Pressure	bar	Depends on Site Piping Layout (Maximum Allowable - 9.4 bar)
		xv	Condensing Temperature	⁰ C	Consult with Engineering Department on Case to Case Basis
	7		Physical Data of Condenser		
		i	Overall Length of Condenser	ft	Contact To Engineering Team
		ii	Shell Diameter	inch	Contact To Engineering Team
		iii	Shell Thickness	mm	Contact To Engineering Team
		iv	Shell Length	mm	Contact To Engineering Team
		v	Material of Construction of Shell	-	Mild Steel
		vi	Material Standard of Shell	-	Refer "MOC" Sheet
		vii	Tube Type/ Nature of Tube Surface	-	Integral Helical Fins on the Outside Surface and Integral Helical Ridges on the Inside Surface
		viii	Tube Length	mm	Refer "HX Details" Sheet
		ix	Tube Diameter	mm	Refer "HX Details" Sheet
		x	Tube Thickness	mm	Refer "HX Details" Sheet
		xi	Material of Construction of Tube	-	Cu
		xii	Material Standard of Tube	-	Refer "MOC" Sheet
		xiii	Water Volume in Condenser	Liter	Refer KCPL Chiller Selection System Software
	8		Water Box Details		
		i	Type	-	Standard - Dish Ends (M.W.Box - Optional)
		ii	Material	-	Mild Steel
		iii	Material Standard	-	Refer "MOC" Sheet
		iv	Nozzle size	NB	Refer KCPL Chiller Selection System Software
		v	End connection	-	Standard - Victaulic Conn. (Flanged Conn. - Optional)
		vi	MOC of Water Side Gasket	-	NAM AF 120
		vii	MOC of Refrigerant Side Gasket	-	NAM AF 159
	9		Accessories Provided		
		i	Pressure Relief Valve	-	Spring Loaded (For Safety Valve Set Pressure Refer ESP)

		ii	Drain/Vent Valves	Inch	Plugged Connection Provided (3/8" NPT)
I			Suction Line		
	1		Design Code	-	ASME B31.3
	2		Isolation Valve	-	Butterfly Valve
	3		Material of Construction	-	Carbon Steel
	4		Material Standard	-	Refer "MOC" Sheet
	5		Angle Valve	-	Provided on Suction Line For Oil Recovery Line
J			Discharge Line		
	1		Design Code	-	ASME B31.3
	2		Isolation Valve	-	Check Valve (NRV)
	3		Material of Construction	-	Carbon Steel
	4		Material Standard	-	Refer "MOC" Sheet
	5		Skin Type Thermowell	-	Provided on Discharge Line For Discharge Temp. Sensor
K			Liquid Line		
	1		Design Code	-	ASME B31.3
	2		Expansion Valve		
		i	Type	-	Electronic Expansion Valve
		ii	Make	-	Refer "Make List" Sheet
		iii	Quantity	Nos.	Contact To Engineering Team
		iv	Sight Glass	-	Inbuilt
		v	Moisture Indicator	-	NA
	3		Filter Drier	-	NA
	4		Material of Construction	-	Copper
	5		Material Standard	-	Refer "MOC" Sheet
L			Desuperheater	-	Not Applicable
M			Economizer	-	Not Applicable
N			Starter and Control Panel		
	1		Panel Enclosure	-	Starter and Control Panel Integrated in Single Fabricated Box
	2		Make	-	Kirloskar Approved Vendor
	3		Material of Enclosure	-	CRCA Sheet
	4		Thickness of Enclosure	mm	Fabricated Enclosure - Load Bearing Member - 2 mm Non-Load Bearing Member - 1.6 mm
	5		Ingress Protection (IP)	-	IP54
	6		Painting Specification		
		i	Paint Type	-	RAL 7035
		ii	Standard	-	Coating as per KCPL Standards
	7		Mounting Arrangement	-	Mounted on Chiller
	8		Type of Starter	-	Inverter Built in Compressor
	9		Type of Isolation	-	Common SFU/SDU with Door Handle
	10		Type of Protection	-	Fuses per Circuit
	11		Switchgear Make	-	Siemens
	12		Electrical and Control Cables	-	Power - PVC Insulated Single Core (Vtg. Grade 1.1 kV) Control- PVC Insulated Single Core, Multicore Cable (Vtg. Grade 1.1 kV) Signal- Shielded Cable
	13		Optional Features		
		i	Phase Indicating Lamps	-	Special-Optional
		ii	Hooter	-	Special-Optional
		iii	Energymeter	-	Special-Optional
O			Controller		
	1		Make	-	Refer "Make List" Sheet
	2		Transmitters	-	NA
	3		Oil Level Switch	-	Not Applicable
	4		Oil Level Failure Trip	-	Not Applicable
	5		LP Switch and Gauge	-	No, Controller Program will Take Care of Low Pressure
	6		HP Switch and Gauge	-	No, Controller Program will Take Care of High Pressure
	7		Chilled Water Flow Failure	-	Yes
	8		Cooling Water Flow Failure	-	Yes
	9		Reverse Rotor Protection	-	No
	10		High/Low Voltage Trip	-	Yes
	11		Low Current Trip (Current Based-Analog)	-	Yes
	12		High Current Trip (Current Based-Analog)	-	Yes
	13		Phase Failure/Reverse Phasing Trip	-	Yes
	14		Earth Fault Trip	-	No
	15		Communication Through RS232/RS485	-	RS485
	16		Display of Microprocessor	-	Yes
	17		Type of Display	-	7" Touch Screen Display
	18		Remote Monitoring Facility	-	Yes
	19		Output to DCS	-	Applicable (Only if RS485 is Available)